

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

JetBrains annual survey insights for engineering leadership

23,262 developers surveyed worldwide

Comprehensive snapshot of languages, tools, AI attitudes, and career trends

Audience: Engineering leaders, CTOs, and Developer Experience teams

Python Closes the Gap: 8 Years of Language Usage Shifts

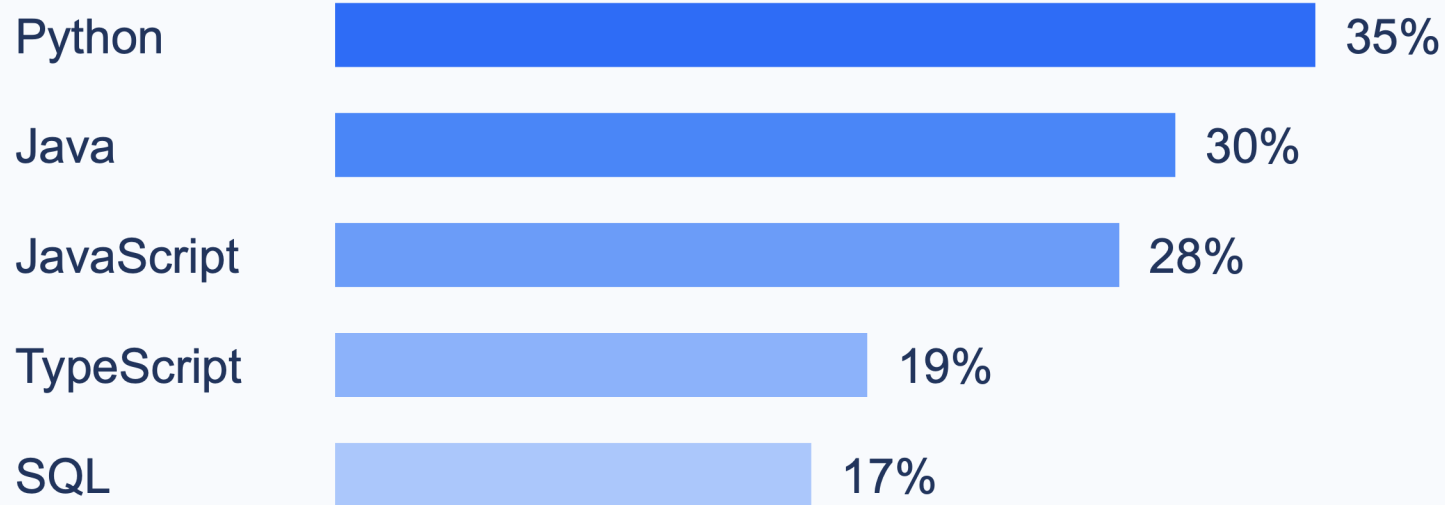
2017-2024 usage (%) for top languages from source brief

Language	2017	2020 Peak	2024	Change	Trend Note
JavaScript	65%	70%	61%	-4	Still #1 overall usage
Python	32%	—	57%	+25	Strongest multi-year growth
Java	68%	—	30%	-38	Large long-term decline
TypeScript	12%	—	37%	+25	Tripled since 2017
Go	8%	—	18%	+10	Adoption plateau after growth
Rust	0%	—	11%	+11	Fast systems-language rise
PHP	30%	—	17%	-13	Continuous decline
Ruby	10%	—	4%	-6	Niche retention only

Narrative: top-5 landscape reshuffles as Python and TypeScript accelerate while legacy web stacks contract.

Python Is Now the #1 Primary Language at 35%

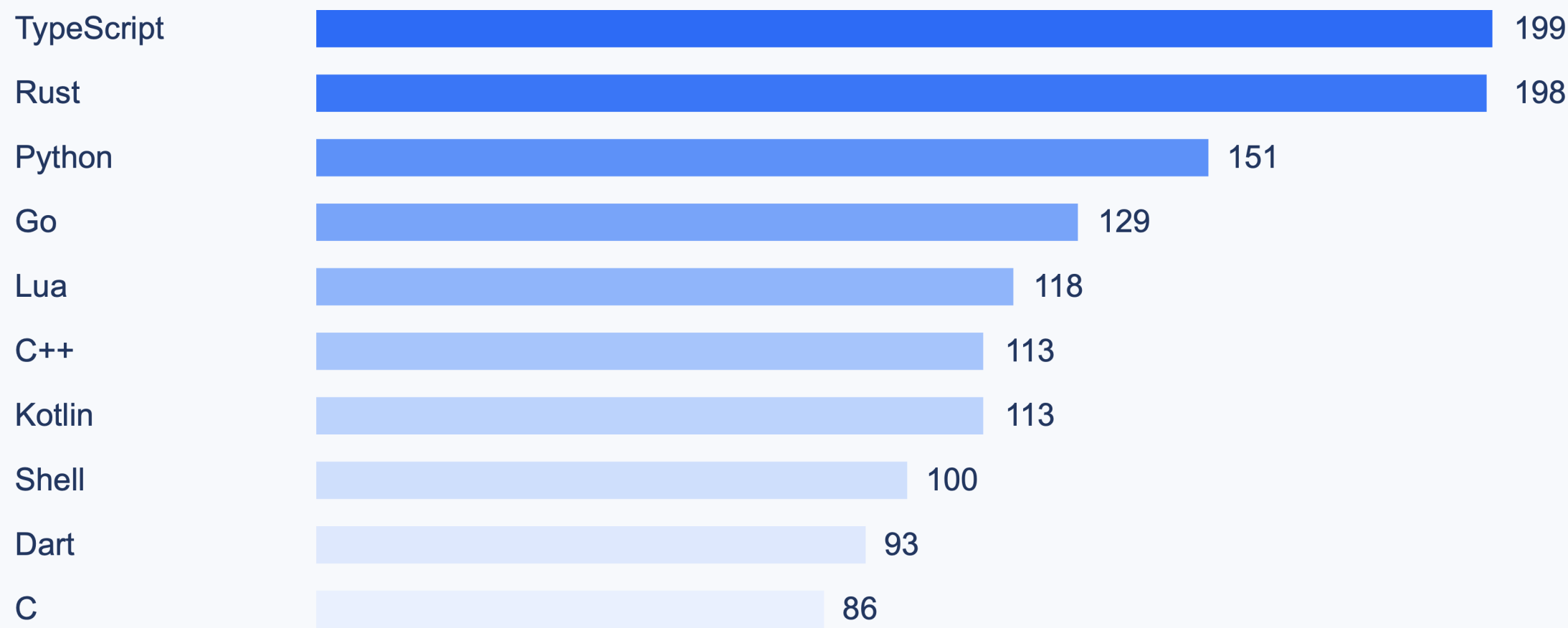
Primary-language distribution vs. adoption intent (<6% across options)



Implication for Leaders

Developers largely remain in existing ecosystems; adoption intent stays below 6%, so hiring strategy should optimize for proven stack depth, not rapid mass migration.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Pattern: type safety and modern tooling are strongly correlated with developer satisfaction.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Science	Python	28%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
User Interface (secondary)	JavaScript	58%
APIs & Services	Java	49%
System Tools & Components	C++	27%
Takeaway: language choice is primarily domain-driven, not a single global preference ranking.		

What This Means for Your Technology Strategy in 2025

- **Invest in Python for AI/ML and data-intensive teams**

Python reaches 57% usage and leads AI/ML (31%) and data science (28%).

- **Standardize on TypeScript for frontend and full-stack UI**

TypeScript has top satisfaction (199) and leads UI workloads at 62%.

- **Build infrastructure bets around Rust and Go**

Go grows to 18%; Rust reaches 11% with near-top satisfaction (198).

- **Plan for shrinking PHP and Ruby talent pipelines**

PHP declines to 17%, Ruby to 4% in long-term usage trends.

- **Use DX and type safety as retention levers**

Top satisfaction languages emphasize modern tooling and strong typing.